

From mPL to Shapley Attribution: Quantifying Per-Cue Location-Privacy Leakage in Images

Technical note — GeoBayes project
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Abstract. A photograph leaks its capture location through many visual cues. We quantify how much each cue leaks with respect to a concrete adversary. Leakage of a cue subset is measured by mPL, a distance-normalized posterior-leakage metric. Because masking effects are non-additive (cues overlap or back each other up), single-cue mPL is a biased per-cue estimate; we treat leakage as a set function and attribute it with the Shapley value, whose efficiency axiom makes per-cue shares sum exactly to the total removable leakage of the identified cue set. The pipeline (GPT-4o cue naming, SAM 3 masks, GeoRanker adversary) is evaluated on 100 im2gps3k images. This revision adds three components. The cue inventory is de-duplicated geometrically before attribution, collapsing 244 named cues into 223 players and raising the median attributed leakage of text/signage from 0.080 to 0.117 without changing the category ranking. The removal operator is upgraded from gray fill to LaMa inpainting, which halves the masking-artifact floor (median control mPL $0.087 \rightarrow 0.045$) and raises the fraction of cues that resolve above their own artifact null from 48.8% to 76.6%, while leaving the ranking and the non-additivity intact. And a second-order anchored Shapley estimator makes the attribution computable from $m + C(m, 2) + 1$ scored subsets instead of 2^m , exactly for $m \leq 3$ and with rank correlation 1.000 beyond it.

1 Problem Statement

Let I be an image with m identified cues $N = \{1, \dots, m\}$, each with a pixel mask, and an adversary who forms a posterior over a fixed candidate set. Two questions: how much belief movement is caused by masking any cue subset (measurement, §2), and how to divide the total leakage among individual cues when effects do not add up (attribution, §4). Two further questions decide whether the answer is an artifact of the measurement rather than a property of the image: who is entitled to name a cue (§6), and what “masking” should physically mean (§7).

2 Privacy Criterion and Adversary Model

2.1 Setup

Let $X \in \mathcal{X}$ denote the capture location of image I , where (X, d_X) is a finite metric space of candidate locations and d_X is the haversine distance. We instantiate X with the $n = 138$ city-level ground-truth labels of our im2gps3k subset (administrative aliases within 2 km merged — only City of Westminster / Greater London, 1.3 km — leaving 137 points; genuinely distinct nearby places are kept separate). For a cue subset $S \subseteq N$, the masked image $I \ominus S$ — removing the union of the SAM 3 masks of the cues in S — plays the role of the sanitized release. Which removal operator implements $\ominus$ is a modelling choice and not a detail: we take LaMa inpainting as the primary operator and gray fill as a robustness variant, and §7 reports what changes between them.

Adversary model. The criterion below is adversary-agnostic; for empirical auditing we instantiate the adversary as a neural posterior estimator with scoring function $s_\theta(x; I)$ — the GeoRanker reward model (Qwen2-VL-7B + LoRA + value head), whose score increases with geographic closeness between the depicted scene and candidate x . Combining the scores by a softmax yields a normalized adversarial belief (we avoid calling it a calibrated Bayesian posterior; see §2.3), with $\tau = 1$ and deterministic decoding at fixed input:

$$\Pr_\theta(x | I) = \frac{\exp(s_\theta(x; I)/\tau)}{\sum_{x' \in \mathcal{X}} \exp(s_\theta(x'; I)/\tau)} \quad (1)$$

2.2 Metric-normalized posterior leakage

Definition 2.1 (mPL, following [4]). For $x_i, x_j \in X$ with $x_i \neq x_j$ and $d_X(x_i, x_j) > 0$, the pairwise leakage of cue subset S is given by equation (2), and the image-level leakage aggregates it over the admissible pair set P (pairs with positive posterior mass under both releases) as in equation (3).

$$\text{mPL}_{i,j}(S) = \frac{1000}{d_x(x_i, x_j)} \left| \log \frac{\Pr_\theta(x_i | I)}{\Pr_\theta(x_j | I)} - \log \frac{\Pr_\theta(x_i | I \ominus S)}{\Pr_\theta(x_j | I \ominus S)} \right| \quad (2)$$

$$\text{mPL}(S) = \frac{1}{|\mathcal{P}|} \sum_{(i,j) \in \mathcal{P}} \text{mPL}_{i,j}(S) \quad (3)$$

Here $\Pr_\theta(\cdot | I)$ is the full-view belief and $\Pr_\theta(\cdot | I \ominus S)$ the counterfactual baseline belief (masked view): mPL measures the distance-normalized odds shift between them; the factor 1000 sets the units to nats per 1000 km. The absolute value makes mPL direction-agnostic. Note $v(N)$, masking all identified cues, is not the total leakage of the image — residual background (climate, terrain, road layout) can still localize; $v(N)$ is the total *removable* leakage of the identified cue set, and §6 shows how much that set depends on who names the cues.

2.3 Calibration and temperature invariance

The GeoRanker score is a ranking/distance reward, not a probability-trained log-likelihood, so the softmax is a belief, not a calibrated posterior. This matters because mPL scales inversely with τ : since s cancels additively in the log-odds,

$$\text{mPL}^{(\tau)}(S) = \frac{1}{\tau} \text{mPL}^{(1)}(S) \quad (4)$$

verified exactly on our data (e.g. $\text{mPL} \times 2.00 / \times 0.50 / \times 0.20$ at $\tau = 0.5 / 2 / 5$). Two consequences. (i) τ is a pure global scale, so within-image subset rankings and interaction signs — on which all our conclusions rest — are exactly τ -invariant; absolute magnitudes are reported in $\tau = 1$ units. (ii) For the absolute scale, temperature scaling can be fit on held-out true-label NLL: on a 47/48 split the optimum is $\tau^* = 0.95$ (near 1), with test NLL $2.21 \rightarrow 2.19$ and Brier $0.643 \rightarrow 0.639$, confirming $\tau = 1$ is already near-optimal. We therefore report a score-induced belief-shift attribution rather than a strictly Bayesian posterior-leakage guarantee.

3 Non-Additivity: Leakage as a Set Function

Define the leakage set function on cue subsets:

$$v(S) = \text{mPL}(S), \quad v(\emptyset) = 0, \quad S \subseteq N \quad (5)$$

Empirically $v(S \cup T) \neq v(S) + v(T)$, in both directions. **Backup** (super-additive): redundant cues compensate for each other, so $v(\{k\}) \approx 0$ for every k while $v(N)$ is large (an Okazaki festival scene shows exactly this). **Overlap** (sub-additive): cues carrying overlapping evidence each cause most of the movement alone, so $v(\{1, 2\}) < v(\{1\}) + v(\{2\})$ (limestone cliffs vs. rocky shoreline in a Capri photo). Consequently no single masking condition yields a fair per-cue number: the leave-one-out marginal $v(N) - v(N \setminus \{k\})$ under-credits backup cues, while the leave-one-in value $v(\{k\})$ over-credits overlapping ones; in our experiments the two disagreed by up to an order of magnitude. Sub-additivity dominates and is not an artifact of the removal operator: under LaMa inpainting 83.8% of the 80 multi-cue images are sub-additive, with median $v(N) / \sum_k v(\{k\}) = 0.621$ (§7).

4 Shapley Attribution

Definition 4.1 (Shapley-mPL). The attributed leakage of cue k is its Shapley value under the set function v , equation (6).

$$\varphi_k = \sum_{S \subseteq N \setminus \{k\}} \frac{|S|! (m - |S| - 1)!}{m!} [v(S \cup \{k\}) - v(S)] \quad (6)$$

φ_k is the average marginal damage of removing cue k over all $m!$ removal orders, and the unique attribution satisfying efficiency, symmetry, null player, and linearity. Efficiency is the property we rely on:

$$\sum_{k=1}^m \varphi_k = v(N) \quad (7)$$

so “cue k contributes $\varphi_k / v(N)$ of the leakage” is well defined. For two-cue images (the median case) the formula collapses to the average of the leave-one-in and leave-one-out readings:

$$\varphi_1 = \frac{1}{2} [v(\{1\}) + (v(N) - v(\{2\}))] \quad (8)$$

Pairwise structure is captured by the Shapley interaction index (SII), which averages the second-order difference over all coalition contexts:

$$I^{\text{SII}}(k, l) = \sum_{S \subseteq N \setminus \{k, l\}} \frac{|S|! (m - |S| - 2)!}{(m - 1)!} [v(S \cup \{k, l\}) - v(S \cup \{k\}) - v(S \cup \{l\}) + v(S)] \quad (9)$$

with $I^{\text{SII}} < 0$ = overlap and > 0 = backup. **Worked example (Capri).** $v(\{\text{cliffs}\}) = 0.088$, $v(\{\text{shoreline}\}) = 0.104$, $v(N) = 0.113$ give $\phi_{\text{cliffs}} = 0.049$, $\phi_{\text{shoreline}} = 0.065$ (sum = $v(N)$), and $I^{\text{SII}} = -0.079$ flags the two coastal cues as overlapping. ϕ_k can legitimately be negative (masking k pushes belief back); such values are reported as measured.

4.1 Second-order anchored Shapley: the computational shortcut

The exact value in (6) needs all 2^m subsets. Where the lattice is not fully scored — the inpainting arm of §7 and the vocabulary ablation of §6 evaluate only singletons, pairs and the grand coalition — we use the second-order truncation of the Harsanyi decomposition. Writing the first- and second-order dividends

$$d_k = v(\{k\}), \quad d_{kl} = v(\{k, l\}) - v(\{k\}) - v(\{l\}) \quad (10)$$

the truncated value and its anchored correction are

$$\phi_k^{(2)} = d_k + \frac{1}{2} \sum_{l \neq k} d_{kl}, \quad \varphi_k = \phi_k^{(2)} + \frac{v(N) - \sum_j \phi_j^{(2)}}{m} \quad (11)$$

The anchoring term restores efficiency by construction, so $\sum_k \varphi_k = v(N)$ still holds exactly. The cost is $m + C(m, 2) + 1$ scored subsets instead of $2^m - 1$ — 16 instead of 31 at $m = 5$, 37 instead of 255 at $m = 8$.

Validation. Against the exact gray lattice on all 80 multi-cue images, the estimator is exact for $m \leq 3$ (31 images with $m = 2$, 30 with $m = 3$; median relative error at machine precision, $< 2 \times 10^{-16}$). At $m = 4$ (18 images) the median magnitude error is 7.7% and the maximum 10.1%; the single $m = 5$ image is off by 54%. Ranking is unaffected throughout: the within-image Spearman correlation between $\phi^{(2)}$ and the exact ϕ has median 1.000 and the top-1 cue agrees on 100% of the 80 images. The median truncation residual $|v(N) - \sum \phi^{(2)}| / |v(N)|$ is 0.031. We therefore read order-2 magnitudes as approximate for $m \geq 4$ and as a ranking only for $m \geq 5$.

5 Experimental Pipeline

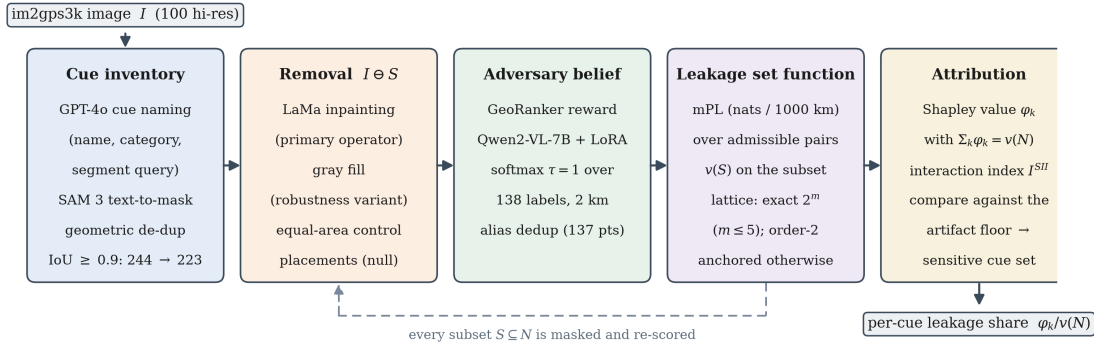


Figure 1: Pipeline. Cues are named by GPT-4o, localized by SAM 3 and merged when their masks coincide; every subset is removed by LaMa inpainting (gray fill as a robustness variant) and scored by the GeoRanker belief over the 138-label metric space; mPL over the subset lattice defines v , which the Shapley value attributes to individual players, and equal-area control placements supply the artifact null a cue must clear to count as sensitive.

Component	Choice
Dataset	im2gps3k subset, 100 hi-res images (95 with maskable cues)
Cue naming	GPT-4o grounded reasoning (name, category, segment query)
Segmentation	SAM 3 text-to-mask; degenerate masks ($>40\%$ of frame) excluded
De-duplication	union-find over mask $\text{IoU} \geq 0.9$; 244 cues $\rightarrow$ 223 players (14 groups, 14 images)
Removal operator	LaMa inpainting (primary); gray fill (robustness variant)
Metric space	138 GT labels + GPS, haversine; 2 km alias dedup (137 points)
Adversary	GeoRanker: Qwen2-VL-7B + LoRA + value head, nf4, deterministic
Elicitation	prompt = image + candidate GPS + label text; softmax $\tau = 1$

Component	Choice
Metric	mPL (nats/1000 km), within-image; full posteriors persisted
Attribution	Shapley ϕ + interaction I^{SII} ; exact 2^m lattice for $m \leq 5$, order-2 anchored otherwise
Artifact null	equal-area control placements: 488 inpainted, 516 gray

Table 1: Implementation summary.

6 Cue Inventory: De-duplication and a Fixed-Vocabulary Ablation

6.1 Geometric de-duplication

GPT-4o names cues in language; SAM 3 grounds them in pixels. Nothing forces two different names to land on different pixels, and frequently they do not: “Thai text on storefront signs” and “Fujifilm signage” are one and the same region of one Bangkok photo. Two players whose masks coincide split a single contribution, and the Shapley value — correctly, given the inputs — pays each of them half. The inventory, not the attribution, is at fault.

We merge cues by union-find over pairwise mask IoU ≥ 0.9 , computed on exactly the per-cue union masks the masking code uses. Containment (recall ≥ 0.95 with IoU below the merge threshold) is reported but never merged — 4 such pairs survive, including the Capri limestone cliffs sitting inside the rocky shoreline. Across the 95 maskable images this collapses **244 named cues into 223 players**, affecting 14 images through 14 merged groups (21 duplicates removed). No re-scoring is required: the members of a merged group cover near-identical pixels, so the masked image of a merged subset equals that of the union of its members’ originals and v' is read straight off the existing complete gray lattice. 12 of the 14 groups are pixel-exact; the remaining two differ by 9.35% and 0.23% of their union area. That is the single approximation introduced by de-duplication.

Credit consolidates as it should: $\phi(\text{merged}) \approx \Sigma \phi(\text{members, before})$, with median $|\Delta| = 0.0063$ and worst case 0.0357 (25.1% of that image’s $v(N)$); the residual is negative in 9 of the 10 partial groups, which is the standard merging effect — a contribution either duplicate could have supplied is now paid once. Efficiency stays exact, $\max |\Sigma \phi_k - v(N)| = 1.11 \times 10^{-16}$ before and after. The effect on the headline is concentrated in one category: the median ϕ of text/signage rises from 0.080 ($n = 29$) to **0.117** ($n = 24$), +45%, because several images (Cambridge, Spa, Bardo, Bangkok, Paris) had two to four separately named signs segmented onto the same pixels. Every other category moves by at most 0.002 and the **category ranking is unchanged**. Pairwise interactions fall from 239 to 192 pairs: 30 before-pairs were pure geometric duplicates, 29 of which had been counted as “overlap” (median $I^{\text{SII}} = -0.058$), so the duplicate-driven overlap disappears with them. The Harsanyi mass moves toward first order accordingly.

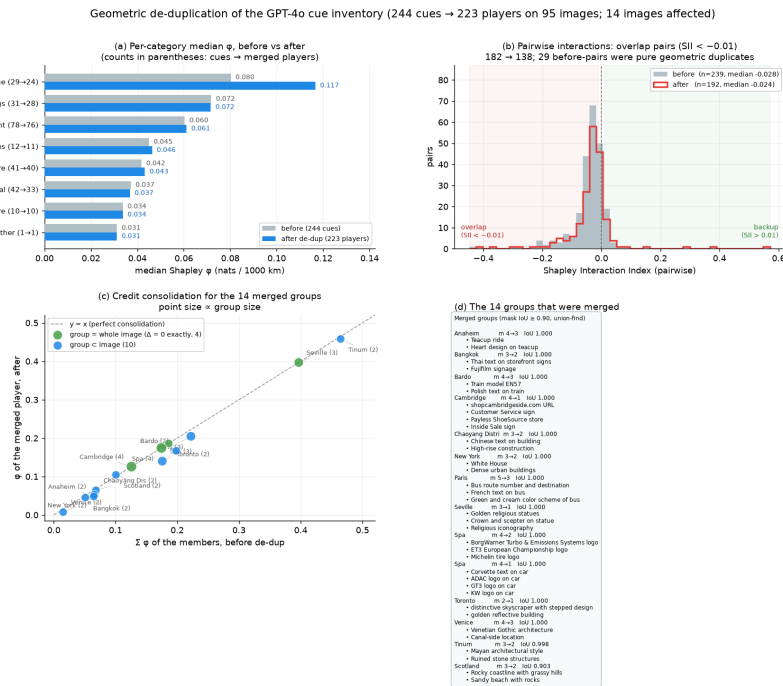


Figure 2: Geometric de-duplication. (a) per-category median ϕ before vs after — only text/signage moves materially; (b) the SII distribution loses its duplicate-driven overlap mass; (c) credit consolidation for the 14 merged groups against $y = x$; (d) the merged groups themselves.

	cue pairs	overlap ($I^{SII} < -0.01$)	$\approx$ add.	backup (> 0.01)	median I^{SII}	Harsanyi order 1/2/3/4 (%)
before de-dup (244 cues)	239	182	34	23	-0.028	60.5 / 32.1 / 13.2 / 3.8
after de-dup (223 players)	192	138	35	19	-0.024	65.5 / 29.7 / 11.6 / 3.2

Table 2: What de-duplication removes. 44 of the 182 overlap pairs vanish, 29 of them pure geometric duplicates. Harsanyi shares are medians of the $|dividend|$ mass per order over multi-cue images (80 before, 76 after; the single $m = 5$ image becomes $m = 3$, so order 5 disappears).

6.2 Who is entitled to name a cue? A fixed-vocabulary ablation

A top-down inventory inherits whatever a language model considers worth mentioning. To bound that dependence we re-ran 10 images with a bottom-up inventory: 12 generic SAM 3 concepts (text or signage, building facade, vehicle, person’s clothing, tree or vegetation, mountain, road surface, street furniture, statue or monument, flag, ...) applied uniformly, with the same removal operator (LaMa inpainting), the same adversary, and the order-2 anchored ϕ of §4.1. Only the definition of “a cue” changes.

Total removable leakage favours the top-down inventory. Summed over the 10 images, $v(N)$ is **4.93** for the 28 GPT-4o cues against **3.50** for the 33 vocabulary cues (0.71 $\times$); GPT-4o wins 8 images, the vocabulary 1 (Bangkok, 1.73 $\times$, where seven generic regions beat three named ones), with 1 exact tie (Seville, where both reduce to the same statue). The median per-image ratio is 0.83 $\times$.

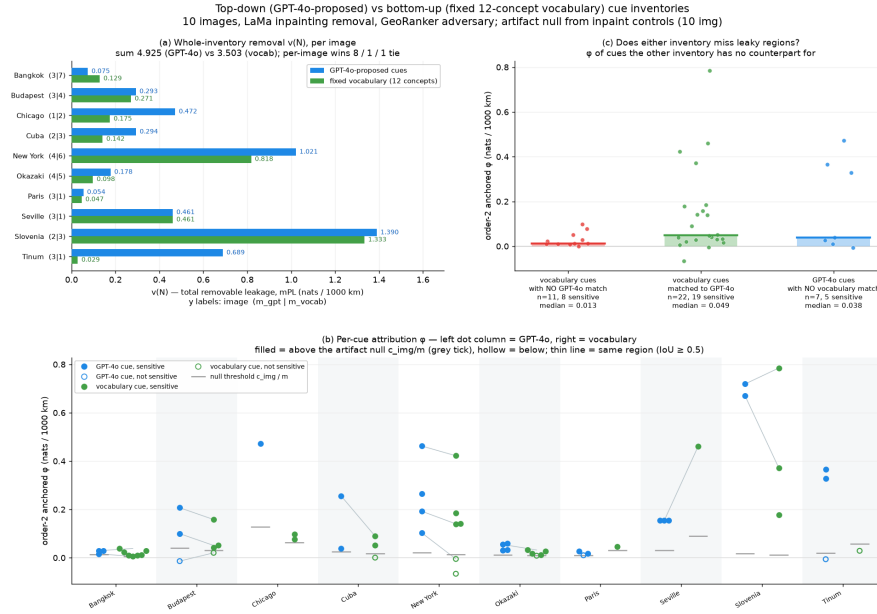


Figure 3: Top-down vs bottom-up cue inventories (10 images, LaMa inpainting, GeoRanker). (a) total removable leakage per image; (b) per-cue ϕ with the per-image artifact null, matched regions joined by a line; (c) the two unmatched populations — the vocabulary’s exclusive finds are numerous but weak, GPT-4o’s are few but strong.

What a fixed vocabulary systematically misses. 7 of the 28 GPT-4o cues have no vocabulary counterpart at $\text{IoU} \geq 0.1$, and 5 of those clear the artifact null: 4 landmarks/buildings, 2 architecture, 1 text/signage. Three gaps are structural rather than accidental. *Masonry and ruins*: at Tinum the vocabulary finds only a 1.5%-area “tree or vegetation” and reaches $v(N) = 0.029$, against 0.689 for “Mayan architectural style” ($\phi = 0.328$) and “Ruined stone structures” ($\phi = 0.366$). *Ornamental architectural detail*: the Paris “Golden clock with ornate design” ($\phi = 0.027$) is not any of the 12 concepts. *Text on portable objects*: the Chicago “Wrigley Field text on cup” ($\phi = 0.472$) carries that image’s entire leakage and no generic concept covers a cup.

The converse gap is real but weak. 11 of the 33 vocabulary cues have no GPT-4o counterpart and 8 of them are sensitive (73%), but their ϕ median is 0.013 against 0.049 for matched cues — mostly generic clothing, road surface and street furniture. Two are not negligible: in Chicago, where GPT-4o named only the cup, “person’s clothing” ($\phi = 0.098$) and “street furniture” ($\phi = 0.077$) are leaky regions the top-down inventory never proposed. Where both inventories do find the same region — 13 mutual-best pairs at $\text{IoU} \geq 0.5$ — the two attributions agree: Spearman $\rho = 0.80$, sensitive-flag agreement 12/13.

Conclusion: neither inventory is a superset. Merging regions across the two sides at $\text{IoU} \geq 0.5$, the union of sensitive regions is **38** = 11 found by both + 11 only by GPT-4o + 16 only by the fixed vocabulary.

The right default is therefore a hybrid inventory — run both and merge geometrically, exactly as §6.1 already does within one inventory. The 95-image results of §8 use the GPT-4o side only and consequently under-count the generic-background leakage the vocabulary exposes; the ordering of the named cues is not affected.

7 Removal Operator and the Masking-Artifact Floor

Gray-filling a mask does not only delete evidence, it also writes a flat gray blob into the image, and a vision model can react to the blob itself. Any belief movement produced that way is a masking artifact, not leakage. We therefore re-ran the entire measurement with LaMa inpainting, which fills the mask with plausible surrounding texture, and read the difference between the two arms as a test of operator dependence: 95 images, 578 scored variants (all singletons, all pairs, the grand coalition) plus 488 equal-area control placements.

(a) The cue ordering survives. Over the 244 paired cues, Spearman $\rho(\text{gray}, \text{inpaint}) = \mathbf{0.856}$ (Pearson 0.879). Inpainting yields the smaller value for 57.4% of cues and the median single-cue mPL falls from **0.098 to 0.092** (median ratio 0.939 $\times$) — a level shift of about 6%, not a reordering.

(b) Non-additivity is operator-independent. Under inpainting 83.8% of the 80 multi-cue images are sub-additive with median $v(N) / \sum_k v(\{k\}) = \mathbf{0.621}$, the same picture §3 draws under gray fill. The empty-context second difference is more negative under inpainting (204 redundant / 18 $\approx$ 0 / 17 synergistic, against the gray SII's 182 / 34 / 23), but that gap is the known bias of the empty-context index — the gray empty-context index gives 215 / 12 / 12 on the same images — not an operator effect.

(c) The artifact floor halves. Each cue's equal-area control masks a region of the same area at a random non-overlapping position; its mPL is the movement the operator produces by itself. Inpainted controls move the belief far less than gray ones: median **0.045 vs 0.087**, P90 **0.131 vs 0.189**. This comparison is *unpaired* — the inpaint control run contains no same-placement gray twins, so the gray side comes from the earlier, independent run at different random positions (516 gray placements against 488 inpainted) and the statement is distribution-level, not per-placement. The consequence is nevertheless large: the fraction of cues whose real removal exceeds the strongest of its own controls — the resolvable fraction — rises from **48.8% to 76.6%** ($n = 244$ each), and the gain is biggest exactly where gray fill was least trustworthy (Table 3). After de-duplication both arms are essentially unchanged: gray 47.1% ($n = 223$), inpaint 76.0% ($n = 221$; two merged players group three originals whose union was never scored in the inpaint arm).

Category	n	resolvable, gray	resolvable, inpaint
road/infrastructure	10	40.0%	100.0%
text/signage	29	65.5%	89.7%
landmarks/buildings	31	51.6%	83.9%
commercial/cultural	42	54.8%	78.6%
vehicles/license plates	12	66.7%	75.0%
environment	78	39.7%	69.2%
architecture	41	43.9%	68.3%
other	1	0.0%	100.0%
all cues	244	48.8%	76.6%

Table 3: Resolvability above the artifact floor (real removal > max of that cue's own equal-area controls). Unpaired: the gray column is the earlier control run at different random placements.

(d) What changes in the attribution. The order-2 anchored ϕ under inpainting correlates with the exact gray ϕ at $\rho = 0.750$ globally, with within-image median 1.000 and 72.5% top-1 agreement; the median ϕ barely moves (0.055 $\rightarrow$ 0.056). One category moves materially: vehicles/license plates, from 0.045 to 0.075. A plausible reading — which we state as a *hypothesis*, not a result — is that gray fill leaves the vehicle's silhouette and its placement in the scene legible, so part of the cue survives the removal it was supposed to undergo, whereas inpainting removes the silhouette too and the measured contribution rises accordingly. Testing it would require a silhouette-preserving inpainting control, which we have not run.

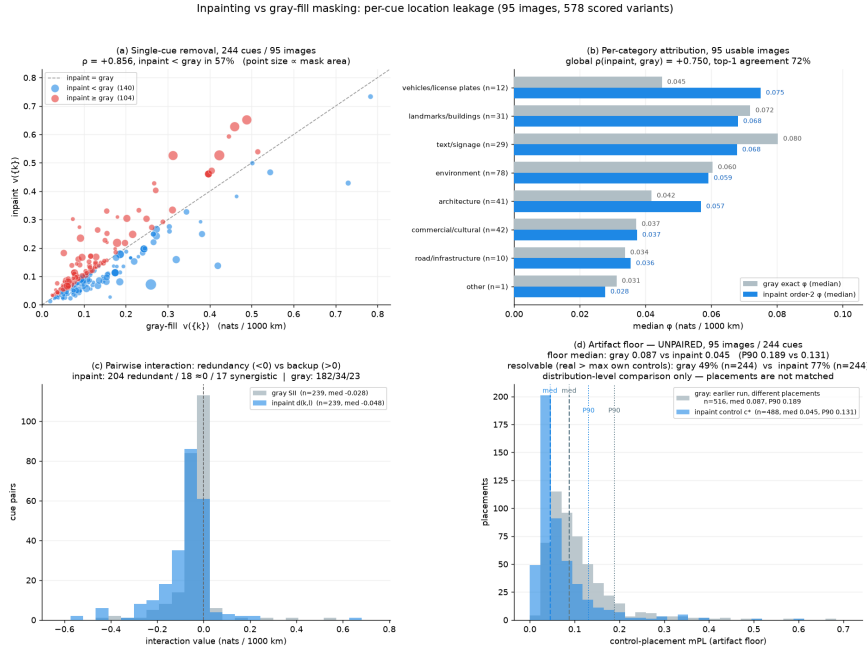


Figure 4: Inpainting vs gray-fill masking. (a) single-cue agreement, $\rho = 0.856$; (b) per-category attribution, gray exact ϕ vs inpaint order-2 ϕ ; (c) the interaction distribution under both operators; (d) the artifact floor and the resolvable fraction — note that the two control populations are unpaired.

8 Results on 100 Images

Across the 95 images with maskable cues the adversary attains a country-scale accuracy (error < 750 km) of 77.9%, 45.3% within 25 km, median error 59 km (median $p_{\text{true}} = 0.137$). Attribution is computed on all 95 images (the primary result) over the de-duplicated inventory of 223 players; we additionally stratify into 71 country-correct and 24 country-incorrect images, since a wrong top-1 does not imply zero belief movement (mPL measures odds shift, not top-1 correctness). The numbers below are the exact gray-lattice values — the only fully evaluated 2^m arm — retained as the operator-robustness variant of the inpainting pipeline of §7, whose within-image georanking they reproduce (median $\rho = 1.000$).

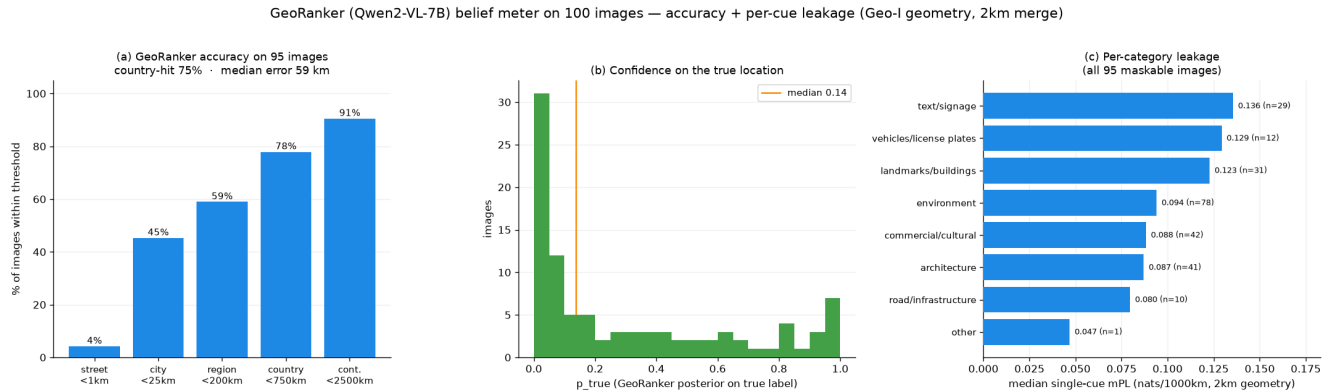


Figure 5: (a) adversary accuracy at increasing error thresholds; (b) confidence on the true label; (c) per-category single-cue mPL (median) on the original 244-cue inventory — the traditional ablation baseline; see Figure 6 for the de-duplicated, Shapley-corrected attribution.

Per-category leakage. As the traditional ablation baseline, the raw single-cue masking effect on the de-duplicated inventory (median mPL, all 95 images) ranks text/signage first (0.133, $n = 24$), then landmarks/buildings 0.120 (28), vehicles/plates 0.111 (11), environment 0.096 (76), architecture 0.086 (40), road/infrastructure 0.080 (10), commercial/cultural 0.074 (33). By §3 this over-states per-cue attribution by double-counting shared leakage. Our primary result is the Shapley-corrected ranking (Figure 6b): **text/signage 0.117** > landmarks/buildings 0.072 > environment 0.061 > vehicles/plates 0.046 > architecture 0.043 > commercial/cultural 0.037 > road/infrastructure 0.034. The correction removes 40–60% of every category’s median but only 12% of text/signage’s, and reorders the middle: vehicles/plates falls from 3rd (raw) to 4th — its apparent standalone importance is largely shared with co-occurring cues in the same scene — whereas text/signage remains 1st by a widening margin, i.e. genuinely the least-shared leakage source. (Restricting to the 71 country-correct images leaves the top three intact but pushes vehicles/plates from 4th to 6th on $n = 6$, confirming that country-conditioning is itself a selection effect; we

therefore keep all 95 as the primary set.)

Shapley attribution. The full 2^m lattice is evaluated for every multi-cue image ($m = 2-5$ before de-duplication, 2-4 after) and augmented with the single-player images ($\varphi = v(\{1\})$, 19 of 95), so all 95 images receive an attribution; the efficiency identity $\sum \varphi_k = v(N)$ holds exactly on the 76 images with $M \geq 2$, to 1.11×10^{-16} . Interactions use the Shapley interaction index of §4 (averaged over coalition contexts), not the empty-context second difference. Across the 192 de-duplicated cue pairs (Table 4, Figure 2b):

Interaction type	Condition	Pairs	Meaning
Overlap (sub-additive)	$I^{SII} < -0.01$	138	either cue alone does most of the movement
$\approx$ Additive	$ I^{SII} \leq 0.01$	35	independent contributions
Backup (super-additive)	$I^{SII} > 0.01$	19	cues must be masked jointly

Table 4: Shapley-interaction structure across the 192 pairs of the de-duplicated inventory (all maskable images; median $I^{SII} = -0.024$). The empty-context second difference over-states overlap on the same pairs (171 / 12 / 9).

Overlap dominance (138/192, 72% by the proper SII) shows that single-cue mPL systematically double-counts shared leakage and therefore overstates per-cue attribution; Shapley redistributes the shared mass so that per-cue contributions sum exactly to $v(N)$. In the 19 backup pairs the single-cue value errs oppositely and understates the contribution. De-duplication removed 44 overlap pairs but did not change this conclusion — 29 of the 44 were pure geometric duplicates, so the remaining overlap is genuine evidential redundancy between distinct regions (§6.1). **Extremes.** Lake Bled is the strongest backup pair — church on an island $\times$ mountainous forested background, $I^{SII} = +0.571$, each cue nearly useless alone — followed by Inca stonework $\times$ vegetation at Cusco (+0.376). The strongest overlap pairs set a built landmark against the vegetation around it: Cuba, stone fortress walls $\times$ lush green landscape (-0.407), and New York, historic building architecture $\times$ trees with autumn foliage (-0.360). Only 2 of the 223 players have $\varphi < -0.01$.

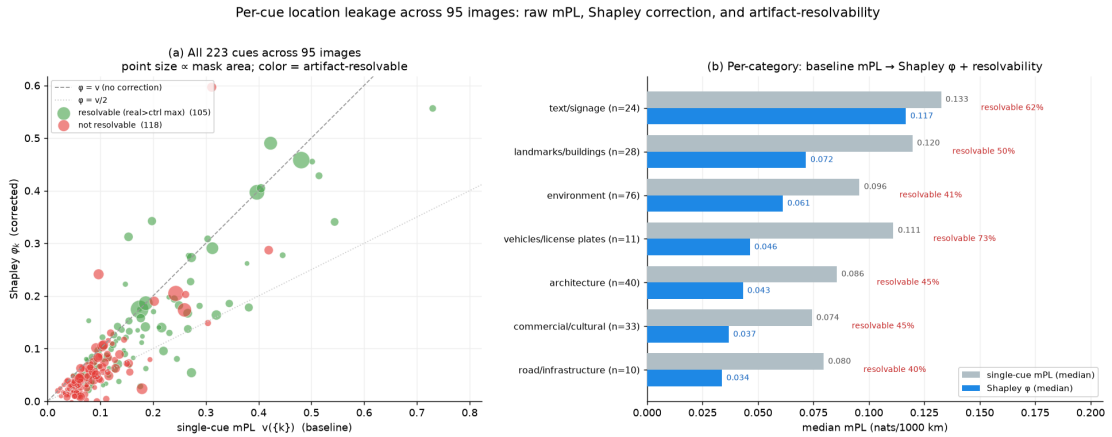


Figure 6: (a) every one of the 223 de-duplicated players, raw single-cue mPL against Shapley φ , sized by mask area and coloured by whether the removal resolves above that cue’s own gray control floor; (b) per-category baseline mPL, Shapley-corrected φ , and the gray-arm resolvable share.

Two images end to end. Figures 7 and 8 show the correction in both directions on single photographs. In the New York scene four cues sum to 1.231 in raw single-cue mPL against $v(N) = 0.823$ — the historic building architecture and the autumn foliage overlap heavily ($I^{SII} = -0.360$) and both are marked down. In the Lake Bled scene the two cues sum to 0.775 raw against $v(N) = 1.346$ — the church and the mountain backdrop back each other up (+0.571), so both are marked up; a defence that removed either one alone would leave the location essentially as identifiable as before.

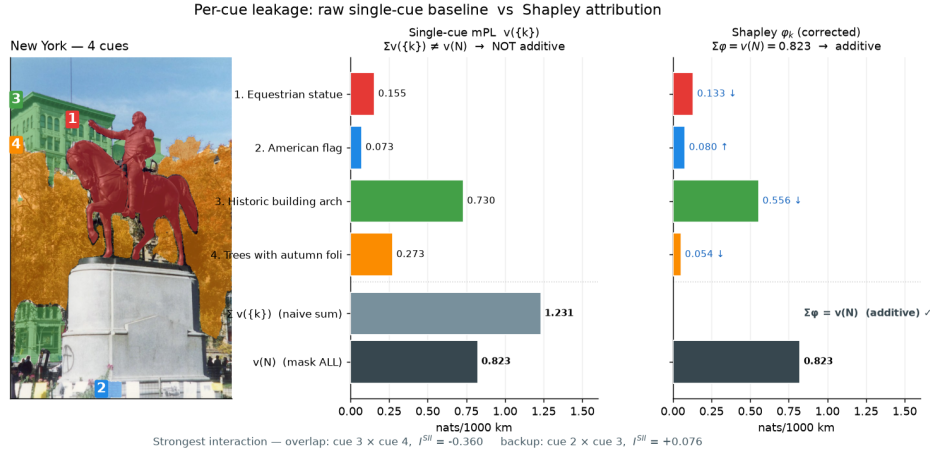


Figure 7: Overlap. New York, 4 cues. $\Sigma v(\{k\}) = 1.231 \neq v(N) = 0.823$; Shapley redistributes the shared mass and the sum becomes exact.

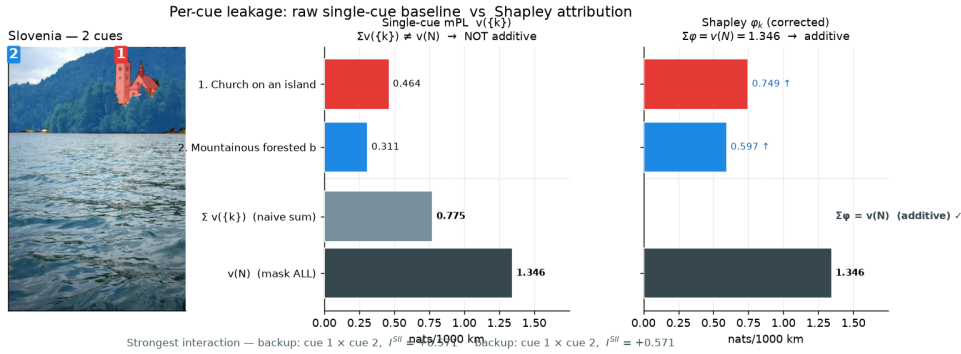


Figure 8: Backup. Lake Bled, Slovenia, 2 cues. $\Sigma v(\{k\}) = 0.775 \neq v(N) = 1.346$; both cues are credited above their standalone readings.

Robustness of the attribution operator. Replacing the Shapley value by the Banzhaf value (uniform coalition weighting instead of permutation weighting) leaves the picture unchanged: Spearman $\rho = 0.984$ across all cues, and the within-image rank correlation has median 1.000. An additive surrogate fitted by least squares to $v(S)$ over all non-empty subsets explains a median R^2 of only 0.668 per image, i.e. the non-additivity that motivates §3 is not a rounding effect. The Harsanyi decomposition puts a median 65.5% of the |dividend| mass at first order after de-duplication, 29.7% at second, 11.6% at third and 3.2% at fourth — pairwise interaction is the dominant non-additive term, which is what makes the order-2 estimator of §4.1 work. All three of these share the same set function $v(S)$ and therefore test the attribution operator, not the measurement; operator dependence of the measurement itself is what §7 tests, and inventory dependence is what §6 tests.

Reproducibility. The 100-image sweep, the full subset lattice, the equal-area controls and the inpainting arm all run locally on the GeoRanker checkpoint with no API calls; only the cue-naming stage uses GPT-4o, and its output is cached per image. Full posteriors are persisted for every scored variant, so the set function, the Shapley values, the interaction indices and the de-duplication can all be recomputed from the stored JSON without re-running the adversary. The prior gray-fill numbers are retained in full as the robustness variant of the inpainting pipeline.

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